Hossein Moosavi Shoa'r

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Thermal-fluid research | Experimental and computational methods

Education

M.Sc. in Energy Conversion 2023–present
Sharif University of Technology, Tehran, Iran
GPA: **19.02/20**. Preparing to defend my M.Sc. thesis.
Thesis: Art of Heat Exchange: Exploring Paint Rheology and Patterns on Heat Exchangers.
Supervisors: Prof. Ali Moosavi and Dr. Hossein Pakzad.

B.Sc. in Mechanical Engineering 2019–2023
Isfahan University of Technology, Isfahan, Iran
GPA: **18.99/20**; ranked **3rd of 120**.
Thesis supervisor: Prof. Ramin Kouhikamali.

Research Experience

Research Assistant — Spray Deposition in Heat Exchangers 2023–present
Micro/Nano Fluidics Laboratory, Sharif University of Technology; thesis research conducted from 2024

- Designed and built three iterations of an experimental rig to study paint penetration and deposition; investigated Newtonian and non-Newtonian paints (DoE)
- Developed paint formulation for repeatable deposition experiments and scanned-sample analysis, addressing adhesion, cleaning and image-contrast limitations.
- Wrote a Python/MATLAB image-analysis pipeline for denoising, colour-threshold segmentation and extraction of twelve coverage, penetration, fragmentation and spatial metrics.
- Benchmarked a two-phase VOF model in ANSYS Fluent and investigated the possible use of Euler–Euler approach

Human Thermal Response and Universal-ODE Modelling Ongoing
Self-initiated research

- Studied human thermal response using the ASHRAE two-node model, with exergy and entropy-generation analysis (Independent Study).
- Integrating Neural networks for human body thermal response prediction; work remains in progress.

Cascade Heat Pumps and Uncertainty Analysis 2025–2026
Conference research — first-author heat-pump study

- Analysed a high-temperature zeotropic cascade heat pump with an ejector and phase separator
- Ran a Monte Carlo analysis to assess the influence of uncertain inputs on thermal-system performance.

Thermoelectric Atmospheric Water Harvesting 2023–2024
B.Sc. research project and subsequent publication, Isfahan University of Technology

- Developed semi-analytical models of thermoelectric-integrated heat exchangers and investigated how flow configuration affects atmospheric water-harvester performance.
- Implemented MATLAB modelling and design calculations; the work resulted in a first-author journal article.
- An application was developed later for general use and design of water harvesters .

Publications

Journal articles

1. **H. Moosavi Shoa'r** and R. Kouhikamali. "Thermo-economic analysis of thermoelectric-based atmospheric water harvesters: Flow configuration impacts." *International Journal of Thermofluids*, 23, 100740, 2024. doi:10.1016/j.ijft.2024.100740
2. B. Rezaee, H. Pakzad, M. Fakhri, **H. Moosavi Shoa'r**, A. Moosavi and M. Aryanpour. "Thermo-responsive nanostructured surface: Beeswax for enhanced condensation performance across solid, liquid, and transition states." *Applied Surface Science Advances*, 32, 100936, 2026. doi:10.1016/j.apsadv.2026.100936

Conference papers — ISME

- "Uncertainty analysis of a high-temperature zeotropic cascade heat pump equipped with an ejector and phase separator." Accepted, 2026.
- "Integral solution of the boundary layer of a power-law fluid over a cylinder considering variable properties and viscous dissipation." Accepted, 2026.
- "Numerical investigation of enhanced cooling of a hot surface using an oscillating impinging jet and nanofluid." Accepted, 2026.
- "Presenting an analytical relationship for evaluating the performance of thermoelectric heat exchangers."
- "Enhancement of dropwise condensation using biocompatible surfaces."

Manuscripts in preparation

- "A phenomenological study of the effect of paint rheology on the penetration of spray paint into porous structures."
- "Optimal thermal conditions for longevity: A thermodynamic approach to ageing and human lifespan enhancement."

Industry Experience

R&D Member 2026–present
Isfahan Casting Industries, Isfahan, Iran

- Work on heat treatment of different parts and perform computer-aided engineering (CAE) analysis for the company.

Engineering Software Project and Apprenticeship 2022–2023
TOMCOV Company; software project supervised by Dr. M. R. Forouzan

- Developed a standalone MATLAB application to size street-sweeper suction channels and fan specifications
- Created CAD models of a flow meter and street sweeper during the 2022 apprenticeship.

Tournaments

Glider Aerodynamic Design 2023
Fly Cup Competition — 6th of 39 teams nationwide

- Wrote a flight-simulation algorithm and varied airfoil selection, wing spacing and fuselage shape to maximise flight distance. Developed the CAD model, used CFD to check the aerodynamic center, and constructed the glider from balsa and foam.

Vehicle Exhaust Heat Recovery 2021
Rahneshan Tournament — 2nd place

- Designed a hexagonal heat exchanger and examined cooling-side surface structures and channel geometries across multiple design points. Calculated thermoelectric power output and performed an economic analysis of the final design.

Technical Skills

Scientific computing: MATLAB (advanced); Python (intermediate); EES (professional); C/C++ (basic).

CFD and numerical tools: ANSYS Fluent (intermediate), meshing and UDF coding; finite-difference solvers; STAR-CCM+, COMSOL and Aspen HTRI (basic).

CAD and fabrication: SolidWorks and CATIA (certified); Inventor; AutoCAD; Arduino programming; laser cutting; lathe operation; welding and visual testing.

Soft skills: Teamwork and leadership developed through competition teams, industrial experience, teaching and volunteer organising roles.

Teaching Experience

Sharif University of Technology, Department of Mechanical Engineering

- Teaching Assistant, Design and Optimisation of Thermal Systems (Spring 2026) — Dr. H. Bahman Jahromi.
- Teaching Assistant, Thermodynamics II (Spring 2026) — Dr. H. Pakzad.
- Teaching Assistant, Power Plant Design (Fall 2025) — Dr. H. Bahman Jahromi.
- Laboratory Instructor, Thermodynamics Laboratory (Fall 2025) — Dr. M. H. Saidi.
- Teaching Assistant, Thermodynamics I (Spring 2025) — Prof. A. Moosavi.
- Laboratory Instructor, Fluid Mechanics Laboratory (Spring and Fall 2025) — Dr. B. Farhanieh.
- Teaching Assistant, Fluid Mechanics I (Fall 2023) — Prof. A. Moosavi.

Isfahan University of Technology, Department of Mechanical Engineering

- Teaching Assistant, Fluid Mechanics II (Spring 2023) — Dr. Saghafian.
- Teaching Assistant, Fluid Mechanics I (Spring 2021 and 2022) — Dr. Madadollahi.
- Teaching Assistant, Mechanics of Solids (Summer 2021) — Dr. Karevan.

Honours and Awards

- Top Scientific Student, Isfahan University of Technology, 2024.
- Direct admission to the M.Sc. programme at Sharif University of Technology through outstanding-student selection, 2023.

Volunteering and Academic Service

- Active member, Red Crescent Organization, Isfahan University of Technology, 2019–2023; helped organise charity festivals.
- Member, Scientific Association of Mechanical Engineering, Isfahan University of Technology, 2023; organised festivals, tournaments and supplementary classes.
- Conference staff, International Conference on Energy Resilience and Sustainability, Tehran, 2024.

Languages

Persian (native); English (high-intermediate; IELTS planned by the end of January 2027).

References

Prof. Ali Moosavi: M.Sc. thesis supervisor, Sharif University of Technology; moosavi@sharif.edu.

Dr. Hossein Pakzad: M.Sc. thesis co-supervisor, Sharif University of Technology; pakzad@sharif.edu.

Prof. Ramin Kouhikamali: B.Sc. thesis supervisor, Isfahan University of Technology; r.kouhikamali@iut.ac.ir.